

the IPS era. It was the first panel that solved the viewing angle and colour reproduction problems that VA and TN suffered from.

- ◆ **Super-IPS (S-IPS), Dual Domain-IPS (DD-IPS) and Advanced Coplanar Electrode (ACE)**- All three are one and the same technology but from different manufacturers: LG.Display, IBM and Samsung respectively. S-IPS laid the technological foundation for the development of the modern IPS panel. It solved the issues of colour shift and gamma shift.
- ◆ **Enhanced or Advanced IPS (E-IPS or AS-IPS)**- This panel type improved the G2G pixel response time to as low as 5 milliseconds and supposedly eradicated the colour shift problem from diagonal viewing angles.
- ◆ **Horizontal IPS (H-IPS) and e-IPS**- The 'horizontal' portion of the name denotes the pixel alignment. In earlier generations of IPS panels, the LC alignment was arrow shaped but in H-IPS it was vertical like a brush head which helped reduce light leakage. e-IPS is a derivative of H-IPS only with wider aperture for light transmission like in modern day panels. It only has a colour depth of 6-bits per even with FRC.
- ◆ **Ultra Horizontal IPS (UH-IPS) or H2-IPS**- An upgraded version of H-IPS with supposedly higher energy efficiency.
- ◆ **Performance or Professional IPS (p-IPS)**- This is again a derivative of H-IPS as it has the same brush head alignment but with a minor yet important difference. It was capable of 8-bit colour reproduction per color using FRC algorithm.
- ◆ **Advanced High-Performance IPS (AH-IPS)**- The introduction of AH-IPS by LG.Displays led to the inception of wide gamut displays with colour reproduction as high as 95 percent of the NTSC gamut. Commonly used in phones, Apple uses AH-IPS in their 'Retina' displays.
- ◆ **Plane to Line Switching (PLS) and Super PLS (S-PLS)**- During the introduction of p-IPS, Samsung launched its own display technology – PLS to rival it and the same is true for AH-IPS whose direct competitor is S-IPS. While the manufacturers labelled their panels differently the core technology remains the same.